THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 2: **Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The degree of the polynomial $y^3 + 2y - 1$ is: 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

2. The value of the linear polynomial $5x - 3$ at $x = 2$ is: 1

(a) 13 **(b)** 12 **(c)** 7 **(d)** -7

3. The value of the linear polynomial $6x - 5$ at $x = -2$ is: 1

(a) -7 **(b)** -17 **(c)** -11 **(d)** 17

4. The value of $1x^2 + 2x - 3$ at $x = 2$ is: 1

(a) 7 **(b)** 2 **(c)** -3 **(d)** 5

5. The sequence $10, 7, 4, 1, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $-3n + 10$ **(b)** $-3n - 7$ **(c)** $-3n + 13$ **(d)** $10n - 3$

6. For the line $y = -3x + 1$, the slope and the y-intercept are respectively: 1

(a) -3 and 1 **(b)** 1 and -3

(c) 3 and 1 **(d)** -3 and -1

7. For the line $y = 5x$, the slope and the y-intercept are respectively: 1

(a) 0 and 5 **(b)** 5 and 0

(c) 6 and 1 **(d)** -5 and 0

8. The point $(1, 8)$ ___ the line $y = 4x + 3$. 1

(a) lies on the line **(b)** is the origin

(c) is the y-intercept of the line **(d)** does not lie on the line

9. A straight line $y = ax$ always passes through: 1

(a) the y-axis at $(0, a)$ **(b)** no fixed point

(c) the point $(1, 1)$ only **(d)** the origin

10. If $y = ax + b$ passes through $(0, 4)$ and $(1, 7)$, then $a + b$ equals: 1

(a) 7 **(b)** 3 **(c)** 4 **(d)** 11

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the values of the linear polynomial $5x - 3$ at $x = 0$ and $x = -1$. 2

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12. Find the values of the linear polynomial $3x + 4$ at $x = 2$ and $x = -2$. 2

13. Find the values of the linear polynomial $-2x + 6$ at $x = 1$ and $x = 4$. 2

14. Find the values of the linear polynomial $4x - 1$ at $x = 3$ and $x = 0$. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A learning platform charges a fixed monthly fee plus a cost per module. For 10 modules the bill is ₹400 and for 14 modules ₹500. Find a and b in $y = ax + b$. 3

16. A gym charges a fixed fee plus a rate per hour of court use: 10 h costs ₹800 and 15 h costs ₹1100. Find the relation $y = ax + b$ and the bill for 20 h. 3

17. Find a and b in °C $= a$ °F$+b$, given that ice melts at 0 °C (32 °F) and water boils at 100 °C (212 °F). Then convert 98.6 °F to °C. 3

OR

The cost of a journey is $C(d) = 100 + 60d$. Find the cost of 15 km and the distance for a cost of ₹700. Is it linear growth or decay? 3

18. A plant 1.75 ft tall grows 0.5 ft every month. Write h in terms of t (months), find the height after 7 months and explain why this is linear growth. 3

OR

A phone bought for ₹10,000 loses ₹800 every year. Find its value after 3 years, write v in terms of t and after how many years it becomes ₹4,000. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Solve: the sum of two numbers is 64 and one is 10 more than the other.
(ii) The ages of a mother and son are in the ratio 3 : 1 and after 5 years their ages sum to 70. Find their present ages. 5

OR

Case study: A water tank has height $h(t) = 3 - 0.5t$ m of water after t months of summer. 5
(i) Make a table for $t = 0$ to 6.
(ii) Is it growth or decay? Why?
(iii) When does the tank become empty?
(iv) Draw (describe) the graph and state its slope and intercept.

20. Find a and b in $^\circ C = a\,^\circ F + b$ given ice melts at 0 °C = 32 °F and water boils at 100 °C = 212 °F. Hence convert 5
(i) 77 °F to °C and
(ii) 25 °C to °F.

OR

A mobile phone costs ₹10,000 and loses ₹800 in value every year. 5
(i) Find its value after 3 years.
(ii) Make a table for $t = 0$ to 8 years.
(iii) Write v in terms of t and explain why this is linear decay.
(iv) After how many years will its value be ₹2,000?

END OF QUESTION PAPER **Total: 40 Marks**